

DELIN QU

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Physical AI ♦ World Models ♦ Vision-Language-Action Models ♦ 1,883 citations, h-index 15

Intelligence begins in imagination, lives in action.

EDUCATION

Fudan University

Ph.D. in Computer Science and Technology

Sep. 2022 – Jun. 2027
Advisor: Prof. Xuelong Li
Shanghai AI Laboratory

Hunan University

B.Eng. in Computer Science and Technology

Sep. 2018 – Jun. 2022
GPA: 3.86/4.00; rank: 1/208

SELECTED PUBLICATIONS

- NVIDIA. **Cosmos 3: Omnimodal World Models for Physical AI**. *arXiv preprint*, 2026; contributor.
- Shalfun Li^{*}, *et al.*, Delin Qu^{*}, ... Qian Wang. **WALL-WM: Carving World Action Modeling at the Event Joints**. *arXiv preprint*, 2026; core contributor.
- Delin Qu^{*}, Haoming Song, Qizhi Chen, *et al.*, Bin Zhao, Dong Wang, Xuelong Li. **EO-1: Interleaved Vision-Text-Action Pretraining for General Robot Control**. *IEEE TPAMI*, 2026.
- Delin Qu^{*}, Haoming Song, Qizhi Chen, *et al.*, Dong Wang, Xuelong Li. **SpatialVLA: Exploring Spatial Representations for Visual-Language-Action Models**. *RSS*, 2025, *Spotlight*; 524 citations.
- Delin Qu^{*}, Jiaru Li, Bin Zhao, Dong Wang, Xuelong Li. **PiX: Pixel Space World Action Model**. *TPAMI*, under review.
- Delin Qu^{*}, Chi Yan, Dong Wang, *et al.*, Bin Zhao, Xuelong Li. **Implicit Event-RGBD Neural SLAM**. *CVPR*, 2024, *Spotlight* (top 2.6%).
- Delin Qu^{*}, Qizhi Chen, *et al.*, Zhigang Wang, Dong Wang, Xuelong Li. **LiveScene: Language Embedding Interactive Radiance Fields for Physical Scene Rendering and Control**. *NeurIPS*, 2024.
- Delin Qu^{*}, Bangyan Liao^{*}, Huiqing Zhang, Omar Ait-Aider, Yizhen Lao. **Fast Rolling Shutter Correction in the Wild**. *TPAMI*, 2023.
- Delin Qu^{*}, Yizhen Lao, *et al.*, Dong Wang, Bin Zhao, Xuelong Li. **Towards Nonlinear-Motion-Aware and Occlusion-Robust Rolling Shutter Correction**. *ICCV*, 2023.
- Bangyan Liao^{*}, Delin Qu^{*}, Yifei Xue, Huiqing Zhang, Yizhen Lao. **Revisiting Rolling Shutter Bundle Adjustment: Toward Accurate and Fast Solution**. *CVPR*, 2023.
- Haoming Song^{*}, Delin Qu^{*†}, *et al.*, Bin Zhao, Dong Wang, Xuelong Li. **Hume: Introducing System-2 Thinking in Visual-Language-Action Models**. *CVPR*, 2026; [†] Project Lead.
- Qizhi Chen^{*}, Delin Qu^{*†}, *et al.*, Dong Wang, Bin Zhao, Xuelong Li. **FreeGaussian: Guidance-Free Controllable 3D Gaussian Splats with Flow Derivatives**. *AAAI*, 2026, *Oral* (top 4.6%); [†] Project Lead.
- Chi Yan^{*}, Delin Qu^{*†}, *et al.*, Zhigang Wang, Dong Wang, Xuelong Li. **GS-SLAM: Dense Visual SLAM with 3D Gaussian Splatting**. *CVPR*, 2024, *Spotlight* (top 2.6%); 710 citations; [†] Project Lead.

WORK EXPERIENCE

Research Intern, Cosmos 3 & Cosmos 3.5

Jun. 2026 – Present

NVIDIA · Advised by Ming-Yu Liu

Santa Clara, USA

- Build unified action spaces for cross-embodiment robot-policy mid-training in Cosmos 3 and Cosmos 3.5.
- Design in-context learning capabilities for generalist robot policies.

Research Intern, WALL-WM

Sep. 2025 – Dec. 2025

X-Square Robot · Advised by Xiaofan Li

Shenzhen, China

- Core contributor to WALL-WM; built pre-training, fine-tuning, and alignment pipelines for world-action models, with real-robot evaluation focused on generalization and deployment robustness.

Research Intern, OpenSS2

Apr. 2025 – Jun. 2025

AgiBot · Advised by Guanghui Ren

Shanghai, China

- Proposed OpenSS2, a hierarchical System-1/System-2 policy using phased flow matching and value-guided test-time selection, improving task success by 24% over π_0 on AgiBot G1.

Joint Ph.D. Researcher, EO-Robotics / IPEC

Aug. 2022 – Present

Shanghai AI Laboratory · Advised by Xuelong Li

Shanghai, China

- Lead end-to-end research on generalist embodied foundation models, from spatially grounded VLA pre-training (SpatialVLA) and interleaved vision-text-action learning (EO-1) to System-2 control (Hume).
- Collaborated with Hugging Face’s LeRobot team to integrate EO-1 into the official LeRobot codebase.
- Drive multimodal data curation, distributed pre-/post-training, and simulation-to-real robot evaluation.
- Mentor junior researchers and interns in project design, implementation, experimentation, and paper writing.

Teaching Assistant

2022 – 2026

Fudan University

Shanghai, China

- COMP130135: Object-Oriented Programming
- COMP620002: Artificial Intelligence

SELECTED PROJECTS

OpenPI Comet

Core Contributor

Led development of Team Comet’s end-to-end $\pi_{0.5}$ training system across multi-dataset pre-training, rejection-sampling post-training, data generation, and evaluation; placed 2nd worldwide in the 2025 BEHAVIOR Challenge, only 0.0085 Q-score behind the winning team.

FastUMI

Core Contributor

Co-developed a scalable, hardware-independent Universal Manipulation Interface for rapid deployment, robust real-world robot data acquisition, and reusable processing workflows.

Any4LeRobot

Core Contributor ♦

1.1k stars

Developed data-conversion, preprocessing, version-migration, and multi-dataset training utilities spanning Open X-Embodiment, AgiBot, RoboMIND, RoboCasa, LIBERO, and RLDS for the LeRobot ecosystem.

BOOK

Bin Zhao, Dong Wang, **Delin Qu**, Haoming Song, Xuelong Li. *Embodied Intelligence: Technology Frontiers and Applications*. Chinese-language book manuscript; under publisher review.

FUNDING & AWARDS

National Natural Science Foundation of China (NSFC)

2024

Learning to Grow: Interactive Developmental Morphology for Embodied Manipulation Policy

- **Question:** What growth-oriented learning and multi-source interaction mechanisms enable embodied manipulation policies to continually improve in real-world environments?
- **Approach:** Develop a spatial-prior-guided action model that continually improves through a closed loop of planning, execution, feedback, and self-exploration.

Tencent Scholarship (top 1‰)

2023

National Scholarship (top 1‰)

2020, 2021

2025 BEHAVIOR Challenge, 2nd Place Worldwide

2025

INVITED TALKS

Real-Time Interactive World Action Models

Jun. 2026

Shanghai AI Laboratory

Shanghai, China

Exploring Spatial Representations for Vision-Language-Action Models

Mar. 2025

Institute of Artificial Intelligence, TeleAI / China Telecom

China

PROFESSIONAL SKILLS

Peer review: CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, TPAMI, AAAI; AAAI Program Committee member.

Physical AI models: VLA models, world models, multimodal pre-training and post-training, in-context learning

Robot learning: Cross-embodiment action spaces, flow matching, diffusion policies, action chunking, test-time computation

Learning systems: Python, PyTorch, Hugging Face Transformers, distributed training, data curation and evaluation

3D perception: Gaussian Splatting, neural rendering, visual SLAM, real-robot evaluation